DreamBeesAI: Reconstructing Multi-User Cloud Computing Environments from Autonomous Agent Infrastructure

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Abstract

As autonomous artificial intelligence coding agents achieve practical utility in software engineering, developers face significant friction in provisioning, isolating, and sharing the execution environments in which these agents operate. Traditional sandboxed execution environments either enforce rigid ephemeral execution models that discard developer state, require complex enterprise container orchestration, or conflate multi-user collaboration with single-cursor screen sharing.

In this paper, we document the design, implementation, and empirical evaluation of **DreamBeesAI (AgentBoxes)**, an open-source, self-hosted system that reconstructs the multi-user time-sharing computer paradigm within a containerized Linux runtime tailored for autonomous agents. We investigate four primary research questions: (1) whether a single containerized Linux operating system can multiplex discrete graphical desktop sessions without state interference, (2) how identity-based sharing models can map onto POSIX permissions and Git attribution without secret leakage, (3) the latency characteristics of in-browser RFB-over-WebSocket transport under concurrent agent execution, and (4) how decoupled compute and persistent volume storage impact operational economics.

Evaluating DreamBeesAI against live Ubuntu 24.04 containers reveals cold provisioning latencies of $2,342 \pm 115\text{ ms}$, in-browser desktop connection latencies of $12.4 \pm 1.2\text{ ms}$ on local interconnects, and non-destructive session suspension in $1,410 \pm 85\text{ ms}$ with 100% workspace file retention. We provide an adversarial analysis of vulnerability vectors—including symlink tarball exfiltration, wildcard Git directory trust, and asynchronous revocation races—and document the architectural remediation for each. Finally, we report honest experimental boundaries: while autonomous engineering tests confirm technical stability, live human usability trials remain formally blocked, and economic analysis demonstrates that commercial cloud hosting of stateful desktops yields prohibitive operational overhead relative to open-source self-hosting.

Keywords: Cloud Desktops, Autonomous Coding Agents, Multi-User Systems, Remote Framebuffer (RFB), Container Security, Bring Your Own Agent (BYOA).

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1. Introduction

Autonomous software engineering agents, exemplified by Anthropic's Claude Code and OpenAI's Codex CLI, have transitioned from conversational autocomplete assistants to active command-line agents capable of self-directed shell execution, dependency installation, multi-file refactoring, test execution, and browser automation [1, 2]. However, executing autonomous agents directly on developer workstations introduces substantial systemic risks:

1. **Unconstrained Host Access:** Agents executing shell commands have access to local environment variables, personal documents, browser profile cookies, SSH private keys, and local network devices.
1. **Environment Contamination:** Agent-driven package installations (`npm install -g`, `pip install`, `apt-get`) can corrupt local toolchains and leave unmanaged background processes.
1. **Collaboration Friction:** Software engineering is inherently collaborative. Existing developer environments isolate execution to a single user's terminal, preventing peers from observing, pairing, or co-directing agent workflows without resorting to high-bandwidth, single-cursor video screen sharing [3].

To address these limitations, cloud-hosted remote development environments have gained widespread adoption, notably GitHub Codespaces [4], Gitpod [5], and Google Cloud Workstations [6]. Simultaneously, specialized agent execution platforms such as E2B [7] have introduced ephemeral micro-virtual machines for short-lived code interpretation. However, existing architectures exhibit a sharp dichotomy: they either provide heavyweight enterprise VMs requiring significant operational management, or ephemeral execution sandboxes that discard workspace state upon process completion.

DreamBeesAI was developed as an experimental investigation into whether the classical **multi-user time-sharing computer** [8] can be reconstructed inside lightweight Linux containers to serve as an accessible, persistent, and collaborative host for AI agents. Under the governing doctrine *"Bring Your Own Agent. We Bring the Box."*, the system decouples agent tooling from cloud hosting, enabling developers to run official agent applications in browser-accessible virtual desktops while maintaining private credentials and shared project state.

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2. Related Work

2.1 Remote Desktops & Web-Based Graphical Interconnects

The Remote Framebuffer (RFB) protocol, popularized by Virtual Network Computing (VNC) [9], provides display-independent graphical transmission over TCP. Traditional VNC implementations require native desktop client installations and expose unencrypted network sockets. Projects such as noVNC [10] modernized this approach by implementing an HTML5 canvas client communicating via WebSockets. Kasm Workspaces [11] demonstrated that browser-native containerized desktop infrastructure could achieve enterprise usability, though primarily focused on single-user application isolation rather than multi-user collaborative workspace engineering. DreamBees builds upon noVNC by introducing an application-level gateway that dynamically multiplexes individual user display ports over a unified HTTP/WebSocket entry point.

2.2 Cloud Development Environments & Decoupled State

GitHub Codespaces [4] established a critical lifecycle pattern for developer workstations: decoupling active compute execution (CPU and memory) from persistent block storage. Pausing a Codespace halts compute billing while preserving uncommitted files and Git branches on persistent disk. DreamBees formalizes this pattern for containerized desktops, demonstrating that graceful container suspension preserves 100% of POSIX filesystem state without VM snapshot overhead.

2.3 Multi-User Operating Systems & Container Namespaces

Unix time-sharing systems [8] pioneered multi-user resource sharing through discretionary access control (DAC) based on user identifiers (UIDs) and group identifiers (GIDs). With the advent of Linux cgroups and namespaces [12], containerization largely pivoted toward single-tenant, single-process microservices [13]. DreamBees deliberately revives multi-user time-sharing *within* a container namespace: running multiple virtual X11 display servers (`DISPLAY :0`, `:1`, `:2`) tied to discrete POSIX users on a single container, thereby achieving multi-user density without VM hypervisor overhead.

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3. Research Questions

This research investigates four core engineering and architectural questions:

- **RQ1 (Graphical Concurrency):** Can a single containerized Linux operating system multiplex multiple concurrent graphical desktop sessions (`DISPLAY :0..N`) without cursor contention, window manager interference, or cross-session display corruption?
- **RQ2 (Multi-User POSIX Integrity):** How can familiar consumer collaboration models (e.g. Google Drive-style owner/editor invitations) be mapped onto POSIX file permissions and Git attribution without leaking private AI agent credentials?
- **RQ3 (Latency & Resource Footprint):** What are the latency baselines (provisioning, connection establishment, frame streaming) and cgroup resource overheads of RFB-over-WebSocket desktop multiplexing compared to dedicated VM solutions?
- **RQ4 (Lifecycle Economics):** Does decoupling active compute from persistent volume storage provide predictable, transparent cost control under heavy, bursty agent compilation workloads?

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4. Methodology & Engineering Process

4.1 The Commissioning Framework

DreamBeesAI was developed under HEAV3NS through fourteen iterative engineering commissions directed by Founding Custodian EGG and Executive Chair MAJO, executed by Deputy Gemini (an advanced agentic AI coding assistant). Each commission operated under explicit adversarial constraints:

- Architectural changes required empirical validation via automated test suites.
- Passing tests were audited adversarially to identify mocked dependencies and synthetic fixtures.
- Architectural pivots (e.g. rejecting separate agent boxes in Commission 009; rejecting screen sharing in Commission 010) were codified into versioned architectural decision records (ADRs).

4.2 Evidence Classification Standard

To maintain strict intellectual and empirical integrity, all findings throughout the project are classified according to a four-tier standard:

- **VERIFIED:** Directly reproduced against live Linux containers, active display bridges, real POSIX filesystems, and cryptographic tokens.
- **SIMULATED:** Verified using mocked external dependencies (e.g., synthetic billing webhooks, offline OAuth callback fixtures).
- **BLOCKED:** External dependencies, live commercial payment rails, or human participant cohorts unavailable in the test environment.
- **FAILED:** Reproducible functional, security, or economic defect.

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5. System Architecture & Implementation

5.1 The Control Plane & Lifecycle State Machine

The DreamBees control plane (`src/provisioning/box-manager.ts`) governs workstations through a 10-state deterministic finite state machine (FSM):

$$\mathcal{S} = \{\text{requested}, \text{provisioning}, \text{ready}, \text{running}, \text{suspended}, \text{stopped}, \text{failed}, \text{expired}, \text{deleting}, \text{deleted}\}$$

Workstation creation requests are dispatched through an asynchronous job ledger (`JobManager`). Storage volumes and isolated bridge networks (`enable_icc=false`) are provisioned via `BoxStorageManager`.

5.2 Independent Multi-User Desktop Multiplexing

To address **RQ1**, DreamBees provisions separate virtual X11 framebuffers inside the container:

- The Primary Owner runs on `DISPLAY :0` (Xvfb + x11vnc on port 5900) as Linux user `operator` (`UID 1000`).
- Invited Collaborators trigger the on-demand creation of `DISPLAY :N` (Xvfb + x11vnc on port `5900 + N`) executing under dedicated Linux accounts `collab_N` (`UID 1000 + N`).

`DisplayGateway` (`src/gateway/display-gateway.ts`) intercepts incoming WebSocket connections at `/api/boxes/:id/display`, verifies the member's cryptographic token and expiration timestamp, and bridges raw TCP RFB bytes to the browser via an internal Python bridge (`/opt/agentbox/display-bridge.py`).

5.3 POSIX Storage Topography & Git Attribution

To address **RQ2**, the storage hierarchy strictly isolates private credentials while permitting collaborative project compilation:

- Private Home Directories (`/home/operator`, `/home/collab_1`): Mode `0700` (`drwx------`). Anthropic credentials (`~/.claude.json`), OpenAI credentials (`~/.codex/config.json`), and individual Git configs (`~/.gitconfig`) are strictly inaccessible to peers.

- Shared Workspace Directory (`/home/operator/workspace`): Mode `2775` (`rwxrwsr-x`), group `agentbox` . The `setgid` bit ensures all files created by any user automatically inherit group `agentbox` , enabling seamless cross-user compilation.
- Scoped Git Trust: Git's `safe.directory` is explicitly configured to `/home/operator/workspace` and `/home//workspace` , *rejecting malicious wildcard repository execution (* `safe.directory " "`).

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6. Experimental Evaluation

6.1 Provisioning & Interconnect Latency Baselines (RQ3)

We conducted latency benchmarks across 15 consecutive provisioning and connection cycles on live Ubuntu 24.04 containers using standard workstation hardware (10 cores, 32GB RAM, NVMe storage).

Milestone	Target SLA	Measured Mean (μ)	Standard Deviation (σ)	Evaluation Status
Cold Provisioning Latency	< 6,000 ms	**2,342 ms**	$\pm 115\text{ ms}$	**VERIFIED**
Desktop RFB Connection	< 100 ms	**12.4 ms**	$\pm 1.2\text{ ms}$	**VERIFIED** (Localhost)
Secondary Display Spawn	< 1,500 ms	**385 ms**	$\pm 28\text{ ms}$	**VERIFIED**
Instant Revocation Drop	< 50 ms	**12.1 ms**	$\pm 0.9\text{ ms}$	**VERIFIED**
Workspace Tarball Export	< 500 ms	**82.3 ms**	$\pm 4.5\text{ ms}$	**VERIFIED**
Suspend / Resume Cycle	< 3,000 ms	**1,410 ms**	$\pm 85\text{ ms}$	**VERIFIED**

Cold provisioning completes in ~2.3 seconds—substantially faster than commercial virtual machine instances (which typically require 30 to 90 seconds). Desktop RFB connection establishment requires ~12ms on localhost interconnects.

6.2 Resource Contention Under Agent Execution (RQ1, RQ3)

To evaluate multi-user stability under bursty agent workloads, two collaborators concurrently ran full TypeScript test suites and Rust compilations within a Power tier container (4 vCPU, 8GB RAM).

- CPU shares allocated via cgroups prevented host starvation, with both user sessions receiving equitable CPU time slicing.
- Virtual display servers (`Xvfb :0` and `:1`) maintained independent render loops without frame tearing or cross-display event bleeding.
- Terminating Collaborator 1 via the revocation API dropped their WebSocket stream and killed active processes in 12.1ms without disturbing Owner 0's active session.

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7. Adversarial Failure Analysis

During iterative development, rigorous adversarial testing uncovered several critical vulnerability vectors that were systematically analyzed and remediated:

7.1 Symlink Traversal Directory Exfiltration

- **Vulnerability:** Pre-013 implementations utilized `tar -chzf` to create downloadable workspace archives. An adversary inside the container could create a symbolic link pointing to host-sensitive container files:

```bash
ln -s /etc/shadow /home/operator/workspace/shadow_leak
```

- **Remediation:** In `src/provisioning/box-driver.ts`, the archive pipeline was replaced with an inline Python walker utilizing `os.path.realpath`. Any symlink resolving outside `/home/operator/workspace` is filtered out of the stream. A parallel check was implemented in `GET /api/boxes/:id/files`.

7.2 Unbounded Git Trust Exploitation

- **Vulnerability:** Early container configurations executed `git config --system --add safe.directory "*"` to suppress Git ownership warnings in the shared directory. This allowed any user to commit malicious Git hooks or submodule configurations that executed in other users' shells.
- **Remediation:** Wildcard trust was revoked. Git configuration was restricted strictly to `/home/operator/workspace` and `/home/*/workspace`.

7.3 Asynchronous WebSocket Revocation Race

- **Vulnerability:** Early revocation logic removed user metadata in the database but failed to sever existing TCP/WebSocket connections, allowing revoked collaborators to maintain active desktop control until their browser tab closed.
- **Remediation:** Active member WebSockets are registered synchronously in `DisplayGateway`. The revocation handler triggers immediate socket termination (`ws.terminate()`), severing the stream in 12.1ms, followed by `pkill -9 -u `.

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8. Limitations & Honest Disclosures

To ensure intellectual defensibility, we explicitly disclose the empirical limitations of this study:

1. **Absence of Live Human Usability Cohorts:** While automated test suites confirm 100% functional and security compliance, the five-person human usability study remains formally **BLOCKED**. We do not present automated headless test results as evidence of human satisfaction.
1. **Localhost Latency vs. Internet WAN Conditions:** The measured 12.4ms RFB connection latency reflects local loopback socket communication. Over wide-area internet connections, latency will be governed by round-trip time (RTT), packet jitter, and bandwidth constraints.

1. **Intentional Shared Workspace Execution:** As documented in the trust model, collaborators intentionally share write access to `/home/operator/workspace`. If an invited collaborator commits a malicious `package.json` script and the owner executes `npm test`, that script executes within the owner's context. DreamBees provides isolation between *unrelated* tenants and protects private *credentials*, but does not isolate collaborators from code intentionally shared in the project repository.

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9. Operational Economics: Why Commercial Hosting Was Declined

A central question of the DreamBees experiment was whether self-service containerized workstations represent a viable commercial cloud business. Following Commission 013.C, the founding custodians concluded that commercial hosting was unviable for an independent lab:

1. **Hyperscaler Margin Squeeze:** Compute and block storage are commodity utilities. Reselling raw compute at $0.05–$0.10/hour yields gross margins under 15% after accounting for credit card processing fees (2.9% + $0.30) and egress network bandwidth.
1. **Support Overhead of Stateful Desktops:** Stateful Linux desktop environments are notoriously fragile. Users frequently corrupt packages, break X11 configurations, or fill disk volumes, creating customer support ticket burdens that exceed subscriber lifetime value.
1. **Abuse Vectors:** Open, self-service Linux compute with automated terminal access is an immediate target for cryptocurrency miners and botnet operators, requiring costly 24/7 security operations.

Consequently, HEAV3NS determined that the optimal disposition of the work was **open-source release under Apache 2.0**, enabling independent developers and organizations to self-host the platform without commercial intermediary markups.

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10. Conclusion

DreamBeesAI demonstrates that the principles of multi-user time-sharing operating systems, when combined with modern containerization and in-browser RFB streaming, provide a robust, lightweight foundation for autonomous AI coding agents. By isolating agent execution inside containerized environments while preserving independent collaborative desktop viewports and shared Git storage, the platform resolves critical security and setup barriers without sacrificing developer familiarity.

The complete software artifact, test suites, and documentation are preserved under the Apache License 2.0 at `https://github.com/CardSorting/agentboxes-dreambees`.

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